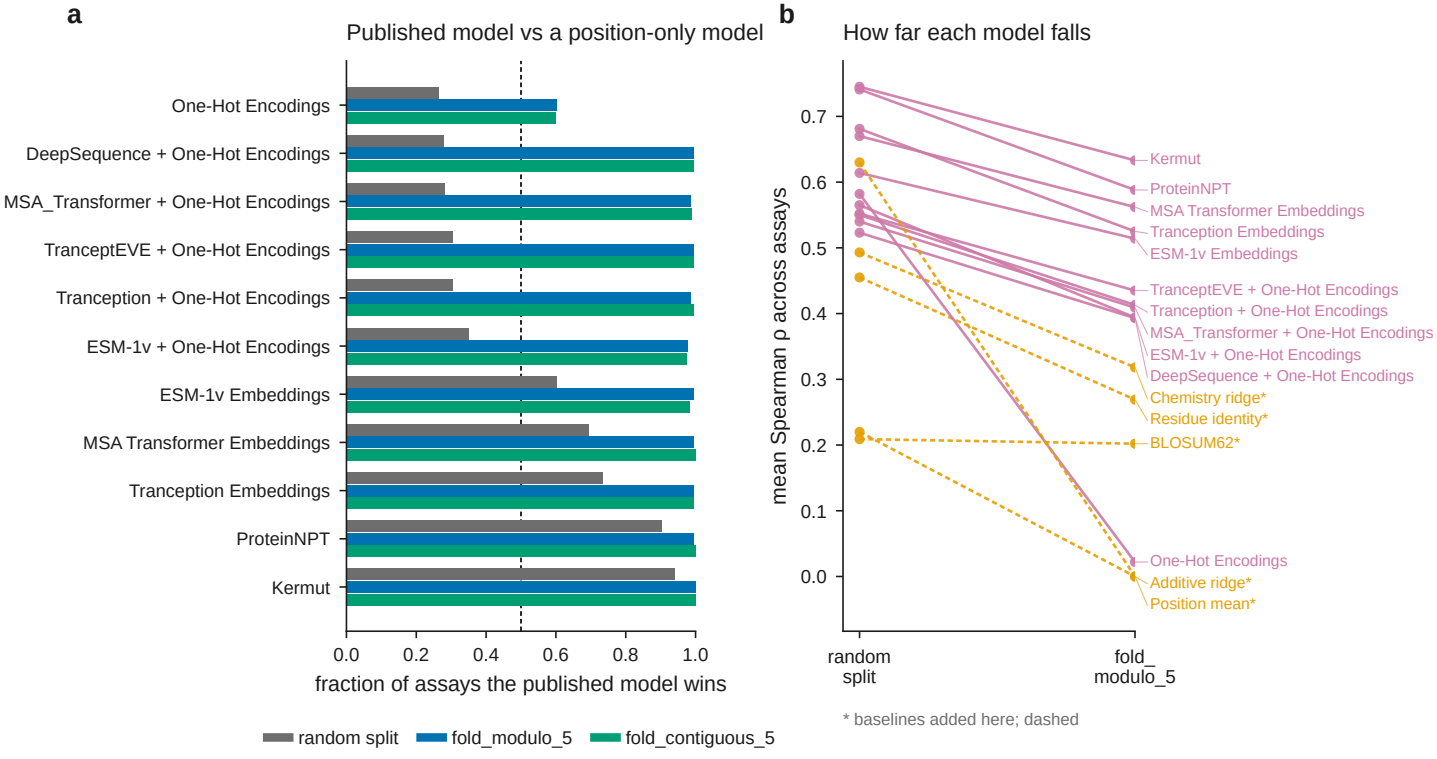
a
Published model vs a position-only model
One-Hot Encodings
DeepSequence + One-Hot Encodings
MSA_Transformer + One-Hot Encodings
TranceptEVE + One-Hot Encodings
Tranception + One-Hot Encodings
ESM-1v + One-Hot Encodings
ESM-1v Embeddings
MSA Transformer Embeddings
Tranception Embeddings
ProteinNPT
Kermut
0.0
0.2
0.4
0.6
0.8
1.0
fraction of assays the published model wins
random split
fold_modulo_5
fold_contiguous_5
b
How far each model falls
mean Spearman ρ across assays
0.0
0.1
0.2
0.3
0.4
0.5
0.6
0.7
Kermut
ProteinNPT
MSA Transformer Embeddings
Tranception Embeddings
ESM-1v Embeddings
TranceptEVE + One-Hot Encodings
Tranception + One-Hot Encodings
MSA_Transformer + One-Hot Encodings
ESM-1v + One-Hot Encodings
DeepSequence + One-Hot Encodings
Chemistry ridge*
Residue identity*
BLOSUM62*
One-Hot Encodings
Additive ridge*
Position mean*
random split
fold_ modulo_5
* baselines added here; dashed